

Chaotic Circuit

T. Nguyen (z5416116)¹

¹UNSW PHYS3117 - Physics Laboratory
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INTRODUCTION

"Given for one instant an intelligence which could comprehend all the forces by which nature is animated and the respective situation of the beings who compose it-an intelligence sufficiently vast to submit these data to analysis-it would embrace in the same formula the movements of the greatest bodies of the universe and those of the lightest atom; for it, nothing would be uncertain and the future, as the past, would be present to its eyes."

- Pierre Simon Laplace, *A philosophical essay on probabilities*, pg. 4 [1]

The topic of determinism has been discussed by philosophers and physicists alike for millenia and will continue for the next thousands of years [2]. In a philosophical sense, determinism is more about free will and moral responsibility of human beings for their actions, whereas the interpretation in physical sciences, relates to predictions given the present state of systems. It is outside the scope of this report to, firstly, argue the necessity for determinism in governing this universe and, secondly, discuss the interplay of free will and determinism, we will leave this to contemporary philosophers and budding online intellectuals.

In the realm of chaos theory, I will assume that determinism is the correct interpretation of our universe, that is, to assume that the demon is real and computationally possible. The definition of determinism I will be abiding by is one given by Hoefer's paper on "Causal Determinism" (2003), [3], where determinism is loosely encompassed by the definition:

Determinism is true of the world if and only if, given a specified way things are at a time t , the way things go thereafter is fixed as a matter of natural law.

In other words, the world this report resides in, is one governed solely by Newton's classical laws of physics. I will ignore the challenges to determinism that quantum mechanics brings [4] and instead focus on the certainty of states given their initial conditions. The study of chaos is, counter-intuitively, not probabilistic. Chaotic systems do not have different outcomes if a perfect process is repeated. Instead, the different outcomes only appear to us humans because of our imperfect knowledge of the parameters of the initial state of the system. Therefore, to an all knowing demon who is able to perceive the parameters of a system to infinite figures, the future and the past, are one and the same.

The language of chaos is often, understandably so, misused as a synonym for randomness. This comes from their shared characteristic of unpredictability however the root cause of this unpredictability is not the same. Chaotic systems are inherently unpredictable due to our lack of mathematical precision and the system's incredible sensitivity to initial conditions. The time evolution of these systems may or may not exhibit patterns to us which also propagates the notion of randomness however they are not equivalent. It is important to clear this language as throughout this report, I may describe trajectories or shapes as disordered and lacking form but this does not mean it is not governed by deterministic physical law.

The conditions for chaos are well understood, being formulated at the time of the discovery of chaotic behaviour by Edward Lorenz in 1963 [5]. The discovery was serendipitous and as the story goes, Lorenz was trying to model weather patterns using the state space of a 12 component model. Each component possessed a differential equation relating to a real world atmospheric parameter such as temperature, pressure etc. He tried to solve this model numerically however found that his model was nonperiodic whereas most studied solutions were periodic.

The famous story in his career is that one day, he wanted to examine one of the solutions in greater detail. He had inputted in 12 numbers that the simulation had printed out, effectively restarting the process from about half way and found that his two runs had completely different outputs despite their shared initial conditions. Later, he would attribute the truncated output to be the source of this divergence [6].

The computer he used for the simulation stored the output numbers to six decimal places however when he printed them out to restart the process, he only printed up to three decimal places to save paper. This one part in ten thousand difference had caused the simulation to wildly alter its course from its original trajectory [7]. The first condition of chaotic motion is then the system must exhibit sensitive dependence on its initial conditions.

The story then goes that while visiting his colleague, Barry Saltzman, he found that in Barry's work, there was a three component subset that displayed aperiodicity, or chaotic motion. This three component model became the famous Lorenz model for which chaotic motion can arise [8]. The beauty of the Lorenz equations came from their simplicity, only requiring three variables for the properties of chaos to appear. Through his work and shown in his paper in 1963, another condition for the aperiodicity is the equations must be non-linear.

Remarkably, the development of chaos theory since its discovery in 1963, continues on the work of Newton in his final pursuit for a solution in the three body problem. Coincidentally, the three body problem has conditions that align with the ones set out by chaos theory. Newton successfully solved the two body problem when he introduced to the world his idea of gravitation. In that scenario, he only needed to worry about the motion of two celestial objects and therefore, the problem could be solved analytically and the solution exhibited periodicity and stability always. However, the inclusion of a third gravitational body would introduce a non-linear term into the equations as well as increasing the variable count to three, satisfying all conditions for aperiodicity and therefore the solution to Newton's three body problem is chaotic [9].

The word 'chaos' was first coined by Jim Yorke to describe types of non-linear dynamical systems however the term has transcended the technical community. The idea of chaos and its child, 'The Butterfly Effect', has become apart of everyday language and integrated itself seamlessly in pop culture. It is common to hear in economics and political discussions, the term 'bifurcation' in relation to causal relationships, especially one where predictability of outcomes is at its lowest. There is a Wikipedia article dedicated to the Butterfly Effect in popular culture as the metaphor of a butterfly flapping its wings to cause a tornado in Texas (presented by Lorenz in a 1972 talk) resonates with the general public and their perception of the world around them. The imagery of the smallest detail influencing the world's most significant events is one of the most powerful narratives to ever been told. But the best kind of stories are the ones that did actually happen and its our marvel of the possibility turning into reality is the reason why the butterfly effect and chaos theory really has a special place in our society today.

THEORY

The central characteristic of chaos is that the system does not repeat its past behaviour, not even approximately. Despite their lack of regularity, the aperiodicity still follow the deterministic equations derived from Newton's laws.

Conditions for chaos

As mentioned before, Edward Lorenz had come up with the two conditions for chaotic motion. For a system to be chaotic, it must:

1. be described by at least three components that are independent dynamical variables
2. contain a non-linear term in its equations of motions that couples several of the variables

The sensitivity to initial conditions is a consequence of the second condition. The non-linear term grows errors in predictions exponentially in time so any small shifts from its original state will cause that state to become essentially unrecognisable in evolution. It is this consequence that will drive the motivation for investigation in chaos theory. Another non-trivial consequence of the second condition is that differential equations containing non-linear terms are impossible to solve analytically and so solutions for equations of motion will have to be done numerically.

Phase space

To study dynamical systems, often phase spaces are used to describe states and their evolution. The phase space is a mathematical space with orthogonal coordinate directions representing each of the variables required to define the state of the system. Commonly, the axis of the phase space will come in position, velocity pairs or analogues. For each solution on a phase space diagram, there will be a corresponding phase trajectory.

There is an extremely non-trivial and significant property of these trajectories, being they are non-crossing. Two trajectories, no matter how close, will never cross each other. This is a consequence of determinism, once a state of the system is known, its time evolution into the future and in the past are known as well. A state in two trajectories would render the system indeterminate and therefore contradict the uniqueness of the determinism on which the motion is formulated upon.

Phase spaces of conservative (constant energy) systems preserve the areas in the space. All the points in a given area of phase space will move in such a way to preserve the area occupied by these same points at some later time. If the system does not conserve energy, the phase area or volume will contract. This is known as a dissipative dynamical system and they introduce a mathematical object known as an attractor. The point is defined as the end state, or fully time evolved state, for all finite states that describes the system.

In order to simplify complicated phase space diagrams, Poincare sections are used. They are constructed by viewing the phase space diagram stroboscopically in such a way that the motion is observed periodically. Therefore, the Poincare section will provide information about the ratio of the strobing and natural frequency of the dynamical motion. If the ratio is rational such that the natural frequency ω_0 is equal to $\frac{p}{q}\omega_s$ where ω_s is the strobing frequency and p and q are integers, then there are q points on the diagram. If the ratio is irrational, then no points are repeated on the plot and so a filled in line of the trajectory will appear.

Lyapunov Exponents

The sensitivity to initial conditions can be measured using a Lyapunov exponent. The exponent, λ , can be calculated using the divergence of two slightly differing initial states, x and $x + \epsilon$. After n iterations, their divergence is characterised approximately as [10]

$$\epsilon(n) \approx \epsilon e^{\lambda n}. \quad (1)$$

The error growth is, aforementioned, characterised as an exponential and the Lyapunov exponent, λ , describes the average rate of divergence. The average needs to be calculated over many initial conditions spread over the trajectory. If λ is negative then the slightly separated trajectories will converge and the evolution is not chaotic but if λ is positive then the evolution is sensitive to initial conditions and is chaotic. To find λ , we consider the one dimensional map given by $x_{n+1} = f(x_n)$. The difference between two initially nearby states after the n th step is

$$f^n(x + \epsilon) - f^n(x) \approx \epsilon e^{n\lambda} \quad (2)$$

We can rearrange for λ ,

$$\lambda \approx \frac{1}{n} \log_e \left[\frac{f^n(x + \epsilon) - f^n(x)}{\epsilon} \right].$$

For small ϵ , the expression inside the brackets reduces down to the derivative of f^n with respect to x , from first principles,

$$\lambda \approx \frac{1}{n} \log_e \left| \frac{df^n}{dx} \right|. \quad (3)$$

Here, we take the derivative through using the chain rule for the n th iterate and take the limit as n goes to infinity,

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log_e |f'(x_i)|$$

Dimensionality

So far I have only considered calculating Lyapunov exponents in one dimension. For higher-dimensional systems, the process of obtaining these exponents is non-trivial. It may be helpful to revisit the idea and motivation behind these exponents. The Lyapunov exponents measure the average rate of divergence between neighbouring trajectories on the attractor. The divergence is measured as the distance between the trajectories and so can only be expressed for an one dimensional comparison. If we want to measure the divergence of a two dimensional trajectory, then we

require two Lyapunov exponents, one for each dimension. The exponents then describe the rate of divergence in uniquely defined, orthogonal directions. Therefore, for n -dimensions, we require n Lyapunov exponents.

We can imagine the a small n -dimensional sphere centered at some point on the attractor and the surface of which consists of phase points from nearby trajectories. The sphere will evolve into an ellipsoid in time with different axes contracting and expanding. The average rates of expansion or contraction are the Lyapunov exponents and for the i th axis, the corresponding exponent is defined as

$$\lambda_i = \lim_{t \rightarrow \infty} \log_e \left[\frac{L_i(t)}{L_i(0)} \right] \quad (4)$$

where $L_i(t)$ is the radius of the ellipsoid along the i th axis. If we consider, in $\mathbb{R}^3$, a small volume in phase space. The volume will change in time as

$$V(t) = V_0 e^{(\lambda_1 + \lambda_2 + \lambda_3)t}. \quad (5)$$

Therefore, the rate of change of volume is found by

$$\frac{dV}{dt} = \sum_{i=1}^3 \lambda_i V(t). \quad (6)$$

We can combine it with the divergence theorem, or Gauss's theorem to obtain a relationship for the sum of the exponents, that is,

$$\frac{dV}{dt} = \int \nabla \cdot \mathbf{F} dV = \sum_{i=1}^3 \lambda_i V(t). \quad (7)$$

I need to talk a step back here and talk about how to actually obtain λ_i because not only would neighbouring points on nearby trajectories involve points along the same dimensional axis, we need to consider points off-axis as well and so our one dimensional view of Lyapunov exponents will no longer function here. I first have to define what I mean by dimension. In this section, I will be talking about *capacity dimension*. Consider a d dimensional line, square, cube analogue with side length, L . We can split this object into N pieces with side length, ϵ . For the n th split, there will be $(2^d)^n$ objects with side length $\frac{L}{2^n}$. We can trivially obtain the number of objects N as equal to the ratio of L and ϵ to the d th power, that is $N(\epsilon) = \left(\frac{L}{\epsilon}\right)^d$. If we rearrange for d and take ϵ to go to zero, or equivalently, $n \rightarrow \infty$, we obtain the definition for capacity dimension,

$$d_C = \lim_{\epsilon \rightarrow 0} \frac{\log_e N(\epsilon)}{\log_e \frac{1}{\epsilon}}. \quad (8)$$

The biggest assumption here is that we can take infinite iterations of the set however in practice we

are limited, computationally, to a finite set that is a subset of the infinite set (an implicit assumption). Experimentally, we can estimate the capacity dimension by using a technique called box counting where a map is covered with boxes of side length ϵ which is varied to obtain multiple values for $N(\epsilon)$. Then the graph of $\log_e N(\epsilon)$ vs $\log_e \frac{1}{\epsilon}$ can be plotted as $\epsilon \rightarrow 0$ and the gradient estimated to be d_C .

However, as you may observe from the methodology, this method of defining dimensionality does not account for the changing density in the regions of the boxes, only counting the number of points inside the box. There are two other dimension quantities I will introduce. First is the correlation dimension, d_C and the second is the information dimension d_I . I will simplify these dimension quantities in just suggesting they account for the density of points within the boxes. We can now define a general definition [11] of dimensions of order q :

$$d^{(q)} = \frac{1}{q-1} \lim_{\epsilon \rightarrow 0} \left(\frac{\log_e \sum_{i=1}^n p_i^q}{\log_e \epsilon} \right), \quad (9)$$

where n is the number of phase elements and p_i is the probability that an attractor point falls in the i th element. It is known that d_C , d_I and d_G correspond to $d^{(0)}$, $d^{(1)}$ and $d^{(2)}$ respectively. To learn more about these dimensional quantities, see [12]. As we will find, many fractals will require many measurements to characterise their complex geometry. However, due to limitations of this report, I cannot go into any further detail with these quantities. They represent different methods/approximations in obtaining the dimensions of an object and are therefore, inherently, non-equivalent.

Attractors

I briefly talked about attractors in phase spaces, giving a broad definition of an attractor being the end state for a finite amount of states that describes the system. Here I will go more into detail about attractors, giving a rigorous mathematical definition. We first should start with the idea of *measure*. In one, two and three dimensions, the measure of a solid body E is referred to as the *length*, *area* and *volume* respectively. We are lucky we do not have to deal with any analytical treatment of measure, instead we can leave that to the mathematicians and their investigation of the *problem of measure* [13]. An attracting set, as defined by Milnor [14] is a set A , $A \in \mathbb{R}^n$ satisfying two conditions:

1. The *basin of attraction*, $B(A)$, consisting of all points whose orbits converge to A , has strictly positive measure.
2. For any closed proper subset $A' \subsetneq A$, the set difference $B(A) \setminus B(A')$ also has strictly positive measure.

The first condition states that the basin of attraction must be physically tangible and continuous. The continuous condition comes from the Lebesgue integral definition of measure, where measure can be calculated as the area under the curve for some function. Therefore a set of discrete points will produce zero measure. If we are working in $\mathbb{R}^2$ and $\mathbb{R}^3$ then our basin of attraction will be defined as an area and volume respectively. The second condition refers to how every point in A will be crucial in creating the set. For an attracting set to be considered an attractor, it must satisfy one more condition,

1. it must contain a dense orbit.

Dense orbits are orbits that either passes through, or gets infinitely close with, every point in the attracting set. This condition is imposed to avoid describing an attractor as the combination of two attracting sets.

There are three nonfractal/nonstrange types of attractors: *fixed point*, *limit cycle* and *torus* attractors. The torus attractor is beyond the scope of this report and chaotic dynamics in general.

The term *strange attractor* arises from attractors that exhibit chaotic behaviour, that is, sensitivity to initial conditions. However not all strange attractors are chaotic. The idea of *strangeness* comes from the geometric structure of the motion. Strange non-chaotic attractors display fractal geometry in their phase space but lack the extreme sensitivity to initial conditions, therefore is not classified as chaotic. We will be focusing on chaotic, strange attractors in this report.

Fractals

I have constantly referred to this strange, non-Euclidean geometry as fractals but I am yet to define it. We know that our Lyapunov exponents describe the average rate of divergence in a certain direction. Aforementioned, negative exponents correspond to convergence, nonchaotic behaviour whereas positive exponents describe the opposite. Here I will define much more simply, Lyapunov exponents describe the 'stretching' or expansion/contraction in different dimensions. For negative exponents, they describe contraction of the orbits whereas positive exponents represents expansion of the orbits. Exponents equal to 0 describe zero contraction or expansion.

Kaplan and Yorke [15] developed a heuristic, mathematically imprecise definition for a *Lyapunov dimension* [16],

$$d_L = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|} \quad (10)$$

where j is the largest integer for which $\sum_{i=1}^j \lambda_i > 0$. The condition for strangeness is then reduced to simply d_L must be a non-integer [17]. It is important to note that for non-area preserving/dissipative systems

in two dimensions, chaos cannot arise as a result of the Poincaré-Bendixson theorem. Therefore we need at least three Lyapunov exponents to observe chaos [18].

Now we can try to understand fractal geometry. A nice analogy would be to think of a crumpled piece of paper in a box. It is a 2D object in a 3D space. The crumpled paper will try to expand (and contract in certain directions) to try to fill the empty space. Of course, our analogy exists in Euclidean space whereas the topic at hand is inherently non-Euclidean but hopefully the imagery helps with the understanding of how fractal geometry comes about.

The term *fractals* comes from *fractional dimension*, coined by Benoit Mandelbrot in 1975 [19]. In Euclidean dimensions, sets can describe ordinary geometric shapes. In 0 dimensional space, sets describe points, for 1 dimensional space, sets describe lines (having only length), in 2 dimensions, sets describe surfaces (length and width), and 3 dimensions, sets describe volumes (having length, width and height). In fractal dimensions, the dimension index can take non-integer values (as seen with the different ways to describe dimensionality), indicating that the set will fill the space. For example, a set with fractal dimension 1.1 will behave like an ordinary line but display some two dimensional thickness. The thickness manifests itself as a self-similarity pattern as you zoom into the 'line' more.

EXPERIMENTAL SETUP

RESULTS AND DISCUSSION

CONCLUSION

ACKNOWLEDGEMENTS

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